LLM Limitations and Rigorous Evaluation in Finance

Lecture 13 — When the Model Sounds Right but Is Wrong

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Outline

Why Calibration Matters in Finance

THE BIGGER PICTURE

A language model that answers with apparent certainty is not necessarily correct. Its expressed confidence—a stated probability, a hedging phrase, or a fluent rationale—may diverge sharply from its empirical accuracy. In finance, a confidently wrong forecast cascades through portfolios, loan books, and regulatory submissions.

The calibration problem: does "80% confident" actually mean "right 80% of the time"?

Five fragilities of LLMs in finance (this lecture)

1. **Calibration** — systematic overconfidence
2. **Temporal leakage** — memorising the future
3. **Stock prediction** — statistical vs. economic value
4. **Evaluation** — beyond classification accuracy

What Calibration Means: ECE and MCE

Perfect calibration: among predictions with confidence p, exactly a fraction p are correct,

$$\Pr(Y = 1 \mid \hat{p} = p) = p \quad \text{for all } p \in [0, 1].$$

UNDER THE HOOD

Bin predictions into M equal-width intervals $B_1, \ldots, B_M$ (typically $M = 10$). Plot mean confidence $\overline{p}(B_m)$ vs. accuracy $\mathrm{acc}(B_m)$ — the *reliability diagram*. Aggregate the gaps:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left| \mathrm{acc}(B_m) - \overline{p}(B_m) \right|, \qquad \mathrm{MCE} = \max_m \left| \mathrm{acc}(B_m) - \overline{p}(B_m) \right|.$$

ECE is a weighted average gap; MCE reports the worst-case bin.

Why MCE matters: a single catastrophic region—say 90–100% confidence predictions that are only 40% accurate—is unacceptable in high-stakes finance even if the *average* looks fine.

Worked Example: ECE of an LLM Credit-Risk Classifier

1,000 default predictions, 5 equal-width bins of width 0.2:

Bin	Interval	$\lvert B_m \rvert$	$\overline{p}(B_m)$	$\mathrm{acc}(B_m)$
1	[0.0, 0.2)	120	0.08	0.11
2	[0.2, 0.4)	180	0.31	0.28
3	[0.4, 0.6)	200	0.50	0.47
4	[0.6, 0.8)	300	0.71	0.55
5	[0.8, 1.0]	200	0.88	0.62

$$\text{ECE} = 0.12(0.03) + 0.18(0.03) + 0.20(0.03) + 0.30(0.16) + 0.20(0.26) = \mathbf{0.115}.$$

Reading the number

An ECE of 11.5% is substantial. Well calibrated at low confidence, but **severely overconfident at the top**: 80–100% confidence predictions are right only **62%** of the time. A credit officer triaging by stated confidence would misallocate effort toward the least reliable cases.

Why LLMs Are Systematically Overconfident

Three reinforcing mechanisms:

1. **RLHF-induced overconfidence.** Human raters prefer confident, fluent answers over hedged ones—even when hedging is more appropriate. The reward signal pushes toward confident expression regardless of knowledge.
2. **Token- vs. claim-level confidence.** Token probabilities compound multiplicatively across a multi-token claim. A model calibrated per token can be wildly miscalibrated per *claim*.
3. **Distributional shift.** Financial data is non-stationary. A model trained on pre-2022 text stays confident on topics that have structurally changed—"it does not know that it does not know."

Documented in finance

ChenGreenGulenZhou2024: LLM return forecasts over-extrapolate recent trends and show optimistic bias—human-like extrapolation/over-optimism that *persists after prompt engineering*.

Measuring and Improving Calibration

Getting confidence scores: log-probs from the API; or, for text-only access, self-consistency (k samples, fraction agreeing) or verbalised confidence—though WANG2023SURVEY show the latter is unstable (70% under one prompt, 95% under another).

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Three post-hoc recalibration methods:

- **Temperature scaling:** $\text{softmax}(\mathbf{z}/T)$, one scalar T fit by NLL. $T > 1$ spreads (cures overconfidence); preserves ranking $\Rightarrow$ AUROC unchanged.
- **Isotonic regression:** non-parametric monotone map; flexible but needs a larger calibration set.
- **Platt scaling:** logistic fit on raw scores; robust on small sets.

Calibration: Conformal Sets and Monitoring

Conformal prediction (coverage-guaranteed sets): output $\{\text{default}, \text{non-default}\}$ when uncertain, with $\Pr(Y \in \mathcal{C}(\mathbf{x})) \geq 1 - \alpha$.

Operational rule

Calibration is distribution-specific. Re-estimate **at least quarterly**— part of model monitoring under SR 11-7.

Memorising the Future

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Evaluating an LLM is not just asking questions and scoring answers. The model may have seen the answers—or the test instances themselves—during pretraining. If the benchmark and the training corpus draw on the same news, filings, and forums, measured “performance” may be **memorisation, not generalisation**. This is the most underappreciated failure mode in LLM finance research.

Canonical trap: “Given this Q2-2021 earnings call, did the stock rise next month?” A model with a Jan-2022 cutoff has *already read* the news, the analyst reactions, and the social commentary on the outcome.

Three leakage channels [ziemke2024temporal]

1. **Direct memorisation** — exact/near-verbatim document seen.
2. **Outcome narratives** — “how the trade played out” retrospectives.
3. **Imbalanced corpora** — textual survivorship bias toward extreme outcomes.

The Stakes Are Empirical, Not Theoretical

Memorisation is measurable

LopezLiraTangZhu2025: LLMs have memorised **exact numerical values** of key economic/financial variables inside their training window—it becomes impossible to distinguish genuine out-of-sample forecasting from recall.

Two confirmations of the magnitude:

- RahimikiaDrinkall2024 pre-train *year-specific* LLMs (2007–2023). Properly time-bounded models **outperform** much larger general-purpose models on trading—so off-the-shelf benchmarks overstate true skill.
- GaoJiangYan2025 introduce the **Lookahead Propensity (LAP)**; a positive LAP–accuracy correlation on stock-return and capex tasks confirms apparent skill is substantially lookahead bias.

Practitioner taxonomy: NoguerAlonso2024 catalogues look-ahead sources (training cutoffs, entity embeddings, backtest design) and mitigations.

Point-in-Time Discipline and Anonymisation

Point-in-time (PIT): the data *actually available* at a historical moment, not later-restated values (cf. CRSP/Compustat snapshots).

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A rigorous LLM protocol demands:

1. **Hard temporal cutoffs** — evaluate $\geq$ 3 months past the confirmed knowledge cutoff (publication-lag safety margin).
2. **Document-level timestamps** — verified publication dates; exclude uncertain provenance.
3. **Contextual anonymisation** — replace company names, tickers, and executives with synthetic identifiers.

Why anonymise

"Will *Apple* report revenue above \$100B in Q4 2022?" is a **recall** question, not a forecast. Replace with "Company X, a consumer

Designing Contamination-Resistant Splits

Random splits are wrong for temporal data. Use **temporal splitting** with a gap Δ:

$$\mathcal{D}_{\text{train}} = \{(x_t, y_t) : t \leq T_1 - \Delta\}, \quad \mathcal{D}_{\text{valid}} = \{(x_t, y_t) : T_1 \leq t \leq T_2 - \Delta\},$$
$$\mathcal{D}_{\text{test}} = \{(x_t, y_t) : t \geq T_2\}.$$

$\Delta \geq$ prediction horizon $h +$ publication-lag buffer.

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Three additional controls:

- **N-gram dedup —** hash 13-gram windows (MinHash LSH) against the training corpus; drop high-overlap instances.
- **Factual probing —** if the model answers facts a non-leaked model could not, suspect contamination.
- **Date-blinding ablation —** a big accuracy drop when temporal markers are removed signals pattern retrieval, not reasoning.

Example: A Contamination Audit

500 earnings calls (2022–23); LLM cutoff Dec 2023; predict EPS beat/miss.

1. **N-gram check:** 34/500 transcripts have ≥ 3 matching 13-grams in Common Crawl $\rightarrow$ flagged and removed.
2. **Anonymise** the remaining 466 (names, tickers, executives).
3. **Compare:** accuracy falls **62%** $\rightarrow$ **54%** on anonymisation $\Rightarrow \approx 8$ pp was entity-level memorisation.

Conclusion

Residual 54% still beats the 50% baseline—but the apparent edge has **shrunk substantially**. Contamination auditing is not cosmetic: it changes conclusions.

The Theoretical Barrier: Efficient Markets

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If LLMs digest news, sentiment, and earnings calls faster than humans, perhaps they find price-relevant signals first. But market efficiency sets a hard limit on how much edge *publicly available text* can contain.

EMH (semi-strong): prices already reflect all public information [FAMA1970EFFICIENT]. An LLM trained on public text is *reconstructing* information already traded on—not extracting novel alpha.

Grossman–Stiglitz: markets cannot be *perfectly* efficient—some mispricing must remain to pay informed traders. The question: can LLMs reach that residual edge before it is arbitraged away? (Where genuine value might remain: see appendix.)

Empirical Evidence: 51–65% Accuracy

Consistent pattern: directional accuracy a few points above the 50% chance level on daily/weekly tasks [XU2024STOCK].

- LOPEZLIRATANG2023: ChatGPT headline sentiment yields significant directional accuracy out-of-sample, scaling with model complexity—but the window was unusually volatile and returns concentrated in some news categories. (The benchmark later critiques target.)
- VIDALSSRN2024: four leading LLMs on 30-day S&P 500 direction — accuracy only **marginally above chance**, below the economic threshold after costs.

Above 62%? Check for these artefacts [ZHAO2025FRONTIERS]

Insufficient train–eval gap · no anonymisation · overlapping windows · gross accuracy without transaction costs. After controls: residual **51–54%**—statistically positive at best, economically negligible after costs.

CFA Institute: durable alpha unlikely in liquid, well-covered markets; possible value in small-cap, EM, or private-information synthesis.

When Simpler Models Outperform LLMs

"More sophisticated" $\neq$ more accurate.

ARIMA(-GARCH)

[BOX1970DISTRIBUTION]

- Principled time-series structure (AR, differencing, MA + variance clustering)
- Few parameters, fast to refit
- No contaminable training corpus

LSTM [HOCHREITER1997LONG]

- Learns longer dependencies
- On *pure price-sequence* prediction, often **matches or beats** LLMs [ZHAO2025FRONTIERS]

Three conditions favouring simpler models

1. **Text irrelevance** — target driven by price/volume dynamics.
2. **Short history** — too little labelled data to justify fine-tuning.
3. **Regime changes** — ARIMA re-estimates cheaply; re-fine-tuning an LLM is an order of magnitude more expensive.

Methodological lesson: a random-walk baseline is not

From Statistical to Economic Metrics

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53% directional accuracy can be worthless after costs; 51% can be valuable if the confident calls land on the big moves. Accuracy is necessary, not sufficient. Financial evaluation must speak in returns, risk, and alpha.

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Sharpe ratio (risk-adjusted return):

$$\mathrm{SR} = \frac{\overline{R}_p - R_f}{\sigma_p}, \qquad \text{annualise} \times \sqrt{252}\ \text{(daily)},\ \sqrt{12}\ \text{(monthly)}.$$

Information ratio (active management):

$$\mathrm{IR} = \frac{\alpha}{\mathrm{TE}} \approx \mathrm{IC} \cdot \sqrt{N}.$$

Factor Alpha: Is the Edge Just Known Risk?

The economic question is not "does the strategy make money?" but "does it make money *beyond* known risk premia?"

Factor alpha: regress excess returns on Fama–French factors,

$$r_{p,t} = \alpha + \beta_{\mathrm{MKT}} r_{\mathrm{MKT},t} + \beta_{\mathrm{SMB}} r_{\mathrm{SMB},t} + \beta_{\mathrm{HML}} r_{\mathrm{HML},t} + \varepsilon_t.$$

The intercept α is the return unexplained by market, size, and value exposure—the genuine skill component.

The sobering finding

Many LLM strategies produce α **indistinguishable from zero**: their apparent returns are repackaged factor exposure, not new information.

Walk-Forward and Expanding-Window Validation

Why not k-fold CV? Shuffling leaks the future into training folds.

- **Walk-forward:** train to t, evaluate $[t, t+h]$, advance; aggregate out-of-sample windows.
- **Expanding-window (anchored):** fix the start, grow the window, re-estimate each step—standard in financial ML [LOPEZDEPRADO2018ADVANCES].

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Both need a **purge period** $\Delta_{\text{purge}} \geq h$ between train and eval, so no observation appears in both via overlapping label windows. At iteration k:

$$\mathcal{D}^{(k)}_{\text{train}} = \{t \leq kh - \Delta_{\text{purge}}\}, \quad \mathcal{D}^{(k)}_{\text{eval}} = \{kh \leq t \leq (k+1)h - 1\}.$$

Cost trade-off: full LLM fine-tuning is expensive to re-estimate, so practitioners widen h—at the price of evaluating a staler model late in

Trading-Grade Evaluation: Costs, Capacity, Drawdown

A strategy can win on paper and fail in practice.

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Transaction costs (turnover τ):

$$c_{\text{round-trip}} = 2\tau\,(\text{half-spread} + \text{market impact}).$$

US large-cap half-spreads 1–3 bps; impact 5–20 bps; daily news rebalancing can hit **10–50 bps/day**, erasing any edge.

Capacity (Almgren–Chriss): temporary impact $\approx \eta\,(q/V)$; profitable at \$10M may be unprofitable at \$1B.

Max drawdown: $\text{MDD} = \max_{t_1<t_2}\left(\frac{P_{t_1}-P_{t_2}}{P_{t_1}}\right)$; Calmar = annual return / MDD.

A complete report should state

Gross & net Sharpe · IR vs. benchmark · MDD & Calmar · turnover

The Silent Failure Mode

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Hallucination—fluent, confident, factually wrong—is undetectable by a reader who lacks the correct answer. A junior analyst handed a hallucinated analysis gets no warning signal: the prose reads exactly as plausibly as accurate content.

Four dangerous categories in finance:

1. **Fabricated citations** — plausible journals/SSRN IDs/regulatory sections that do not exist [BANG2023MULTITASK]. Serious liability in compliance filings.
2. **Incorrect numerical facts** — revenue, EPS, EBITDA, spreads. KANG2023HALLUCINATION: GPT-4 class models err on a non-trivial fraction, with *no* lexical signal distinguishing right from wrong.
3. **Phantom companies/executives** — plausible but non-existent entities; material risk in due diligence and KYC.
4. **Anachronistic facts** — stale rates/ratings/leadership stated confidently

Grounding Strategies

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Reduce reliance on *parametric memory*; anchor to verified external sources.

- **RAG** [lewis2020rag] — retrieve verified docs into context. Effectiveness hinges on retrieval quality; topically-relevant but factually tangential context still yields hallucination.
- **Retrieval verification** — FActScore [min2023factscore] decomposes output into atomic facts and checks each against a source (EDGAR, Bloomberg, a regulatory DB).
- **Tool use / function calling** [schick2023toolformer] — fetch live data; model becomes a reasoning engine, not a knowledge store.
- **Self-consistency** [wang2022selfconsistency, manakul2023selfcheckgpt] — facts stable across samples are likelier correct; variable facts flagged for review.

Grounding reduces but does not eliminate

Hallucination Benchmarks for Finance

Benchmark	What it tests	Grounding
FinanceBench	QA over real SEC filings	retrieval vs. synthesis
FinQA	numerical reasoning over tables	program execution
Kang et al.	revenue/EPS/balance-sheet recall	real vs. fictitious firms

- **FinanceBench** [ZHANG2024FINANCEBENCH]: high hallucination rates even for frontier models on multi-step numerical questions.
- **FinQA** [CHEN2021FINQA]: evaluation via *program execution* catches numerical hallucinations a string match would miss.
- KANG2023HALLUCINATION: models confidently answer about *fictitious* companies with invented-but-plausible figures.

Caveat

Benchmark scores $\neq$ real-world reliability. Evaluate on *your* task type, information source, and output format—not aggregate leaderboard numbers.

Lecture 13 Summary

1. **Calibration** — LLMs overconfident; ECE can exceed 10%. Recalibrate (temperature/isotonic/Platt); use conformal sets for high stakes.
2. **Temporal leakage** — in/near-cutoff evaluation inflates results. Hard splits, purge, anonymisation, n-gram dedup. Audits cut accuracy by $\geq$8 pp.
3. **Stock prediction** — 51–65%, upper end from artefacts. After controls, residual alpha is economically modest; ARIMA/LSTM often win.
4. **Economic evaluation** — Sharpe, IR, factor alpha; costs, capacity, drawdown. Accuracy alone is not enough.
5. **Hallucination** — fabricated citations/numbers/entities, no lexical tell. Grounding helps; human review remains necessary.

The disciplined stance

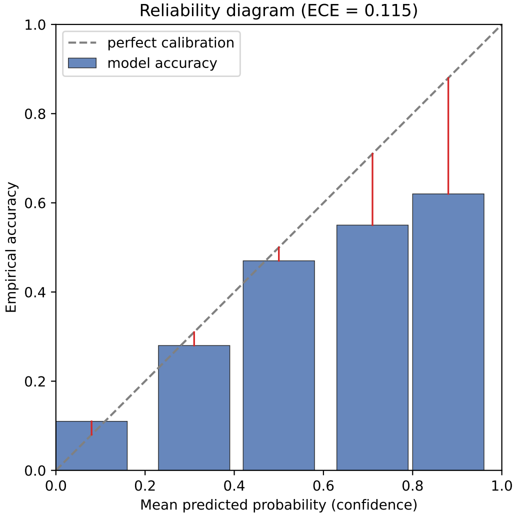
Limitations do not preclude LLMs in finance—they demand treating outputs as **inputs to a decision process**, not end products.

Appendix

Supporting Derivations, Benchmarks, and Further Reading

Extra / optional material — beyond the core lecture.

Appendix: Reliability Diagram (Credit Example)



Bars = empirical accuracy vs. mean predicted probability per bin; dashed diagonal = perfect calibration; red segments = per-bin gaps that ECE aggregates (ECE = 0.115). Well calibrated at low confidence, increasingly overconfident at the top. Generated

Appendix: Adaptive Binning and MCE Detail

Bin-boundary sensitivity. Equal-width ECE is sensitive to boundaries. **Adaptive binning**—equal mass per bin—is more reliable when confidence scores cluster near 0 or 1, as they do for overconfident models [niculescu2005predicting].

When to prefer MCE over ECE. In high-stakes regimes a single catastrophically miscalibrated region (e.g. 90–100% confidence at 40% accuracy) is unacceptable even with a good average ECE. Report both.

Surfacing overconfidence

Manifestations in practice: EPS to two decimals with no interval; definitive credit calls near the boundary; cherry-picked market rationales; "in-/not in compliance" that hides jurisdictional ambiguity.

Appendix: Where Genuine LLM Value Might Remain

Market efficiency limits the edge in *public* text, but residual value can survive in three places:

- **Synthesis at scale** — diffuse signals across thousands of low-attention documents.
- **Novel reasoning chains** — cross-domain links a factor model misses.
- **Private/proprietary text** — not in training, not in prices (raises regulatory questions).

These are precisely the cases the lecture flags as the most defensible uses of LLMs for prediction—small-cap, emerging-market, or private-information synthesis, where public-text efficiency bites least.

Appendix: The Fundamental Law of Active Management

$$\mathrm{IR} \approx \mathrm{IC} \cdot \sqrt{N}.$$

- **IC** (Information Coefficient): cross-sectional rank correlation between predicted and realised returns—*signal quality*.
- **N**: number of independent bets per year—*breadth*, driven by securities covered $\times$ prediction frequency.

Why this matters for LLMs. It separates the two value sources: a weak signal (low IC) can still produce a respectable IR if breadth N is large (many names, high frequency)—but breadth multiplies turnover, and turnover multiplies transaction costs. The decomposition makes the cost/breadth tension explicit.

Appendix: Further Reading

Calibration. GUO2017CALIBRATION (temperature scaling); NICULESCU2005PREDICTING (Platt/isotonic); ANGELOPOULOS2022GENTLE (conformal).

Temporal leakage. ZIEMKE2024TEMPORAL; LOPEZLIRATANGZHU2025 (numerical memorisation); NOGUERALONSO2024 (taxonomy); RAHIMIKIADRINKALL2024 (year-specific LLMs); GAOJIANGYAN2025 (LAP); LOPEZDEPRADO2018ADVANCES (purged CV).

Markets & prediction. FAMA1970EFFICIENT, FAMA1991EFFICIENT; GROSSMAN1980IMPOSSIBILITY; LOPEZLIRATANG2023; VIDALSSRN2024; CHENGREENGULENZHOU2024; XU2024STOCK; ZHAO2025FRONTIERS.

Economic evaluation. SHARPE1994SHARPE; GRINOLD1999ACTIVE.

Hallucination. JI2023SURVEY; KANG2023HALLUCINATION; MIN2023FACTSCORE; MANAKUL2023SELFCHECKGPT; LEWIS2020RAG.